

Emrul Kayes

Staff Software Engineer — C# / .NET Backend, Architecture & DevOps

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PROFESSIONAL SUMMARY

Staff Software Engineer with 10 years building and running backend systems that move money — mobile banking, payment processing, microfinance, remittance, and real-time trading. Deep in the C# / .NET ecosystem: ASP.NET Core APIs, microservices, and the databases behind them. Designs around Clean Architecture, Domain-Driven Design, and event-driven patterns; built SSO from the ground up with IdentityServer4 / OpenID Connect and kept GDPR-regulated data safe for Nordic clients — tightened design and review practices to cut bugs by 30% on a key project. Leads a team day to day while staying hands-on in the critical paths. Open to full-time, consulting, and freelance engagements (4.9/5 average Upwork rating).

TECHNICAL SKILLS

Backend & Frameworks: C#, ASP.NET Core, Web API, Blazor Server/WASM, EF Core, Dapper, SignalR, WPF, Node.js/NestJS	Architecture & Patterns: Microservices, Modular Monolith, Clean Architecture, DDD, Event-Driven, CQRS/MediatR, Vertical Slice, TDD
Cloud & DevOps: Azure (Functions, AD, Storage, Key Vault), Redis Cache, App Insights, Azure DevOps, GitHub Actions, CI/CD, ELK, Kafka	Databases: MSSQL, PostgreSQL, MySQL, EventStore DB, Redis, Relational DB Design
Security & Identity: OAuth2, OpenID Connect, IdentityServer4, Single Sign-On, RBAC, GDPR Compliance	Frontend: Angular, React, Next.js, TypeScript, Redux

PROFESSIONAL EXPERIENCE

Staff Software Engineer, Cefalo Bangladesh Ltd.	<i>Jul 2025 — Present</i>
- Technical lead for a Norwegian payment-platform client — own both team delivery and product architecture; break down features, distribute work, review designs and code, stay hands-on in critical paths. - Design and build ASP.NET Core Web APIs in a microservice architecture, plus Blazor Server/WASM web clients and a WPF desktop terminal. - Own large-volume data handling across PostgreSQL and MSSQL, from schema design through query optimization. - Guard GDPR compliance across the platform — data retention, access boundaries, and audit trails for a privacy-sensitive Nordic market.	
Senior Software Engineer, Cefalo Bangladesh Ltd.	<i>Jul 2022 — Jun 2025</i>
- Built and maintained backend services for SPENN, a multi-country mobile banking and wallet product — transfers, lending flows, and blockchain-connected money movement. - Designed complex microservice architectures, including Azure Function apps for asynchronous and scheduled workloads. - Delivered event-driven systems on ASP.NET Core, applying DDD and established patterns to keep domains cleanly separated. - Built a payroll platform integrating with the HR system via Kafka service-bus messaging, with Google SSO and role-based access control.	
Software Developer, ASAI Management Services Ltd.	<i>Jun 2021 — Jun 2022</i>
- Built the authentication/authorization server from the ground up with IdentityServer4, delivering SSO across the whole product suite. - Designed the application architecture on DDD and Clean Architecture — layered, testable, explicit about its domain.	

- Developed ASP.NET Core APIs backing loan origination: onboarding, KYC verification, disbursement, and repayment scheduling.
- Introduced Redis caching and ELK-based logging to keep the platform observable and fast under load.

Software Developer, BRAC IT Services Ltd.

Apr 2019 — Jun 2021

- Built EndToEnd, a product-lifecycle system that issues, tracks, and destroys banking products such as cards and chequebooks with a full audit trail.
- Developed the remittance management system handling inbound money transfers for BRAC Bank.
- Translated complex business requirements into relational database designs, complex SQL, and stored procedures on N-tier, service-oriented ASP.NET Core / MVC / EF applications.

Software Developer, Dimik Infotech Ltd.

Jul 2016 — Mar 2019

- Built real-time trading features with ASP.NET MVC and SignalR — a live market price board pushing updates to every connected client.
- Implemented portfolio management: deposits, withdrawals, and real-time position tracking for brokerage clients.
- Designed databases and wrote heavily optimized SQL queries and procedures for market-data workloads.

Programmer, Q-Soft Precise Assistance

Jan 2016 — Jun 2016

- Developed and maintained ASP.NET web applications, designed database schemas, and wrote stored procedures.

SELECTED PROJECTS

Convene — MPT & LEAP · Nordic payment & administration platform, Cefalo

Self-service check-in/payment terminals plus the management platform behind them — microservice Web APIs, real-time SignalR flows, and a data layer across PostgreSQL and MSSQL under strict GDPR.

[ASP.NET Core](#) · [Blazor WASM/Server](#) · [WPF](#) · [SignalR](#) · [PostgreSQL](#) · [MSSQL](#) · [OAuth2/OIDC](#) · [Azure](#)

SPENN / Blockbonds · Mobile banking & blockchain wallet, Cefalo

Multi-country mobile banking — wallet-to-wallet transfers, lending, and the integration layer between banking services and a blockchain-connected ledger.

[ASP.NET Core](#) · [Microservices](#) · [EF Core](#) · [MSSQL](#) · [OAuth2](#) · [SSO](#) · [Azure](#)

AMBS · Microfinance & banking, ASAI

Socially responsible microloans across Asia and Africa — customer creation, KYC, disbursement, and repayment scheduling on DDD/Clean Architecture, with an IdentityServer4 SSO server securing the suite.

[ASP.NET Core](#) · [IdentityServer4](#) · [Microservices](#) · [Redis](#) · [ELK](#) · [MSSQL](#) · [DDD](#)

Cefalo Payroll · Internal payroll platform, Cefalo

Company payroll synchronizing HR data over Kafka, Google SSO auth, RBAC-gated actions, and automated salary changes and notifications on promotion.

[NestJS](#) · [Kafka](#) · [MySQL](#) · [TypeORM](#) · [Next.js](#) · [Google SSO](#) · [RBAC](#)

BASE · Stock trading & broker-house suite, Dimik

Broker-house platform for the Dhaka stock market — real-time trading over SignalR, portfolio deposits/withdrawals, and trade execution across web and desktop.

[ASP.NET MVC](#) · [SignalR](#) · [ADO.NET](#) · [EF](#) · [MSSQL](#)

OPEN SOURCE

Quizzer · github.com/mdemrulkayes/Quizzer

Quiz & exam platform as a Modular Monolith (Identity, Quiz, Exam modules) with CQRS via MediatR over a layered Core/Application/Infrastructure/Endpoints structure — JWT with refresh tokens, multi-role RBAC, and timed exam delivery with auto-submit and analytics; Angular signals-based SPA front end.

[.NET 10](#) · [ASP.NET Core Web API](#) · [CQRS/MediatR](#) · [EF Core](#) · [SQL Server](#) · [JWT/RBAC](#) · [Angular](#) · [PrimeNG](#) · [Tailwind CSS](#) · [Docker Compose](#) · [xUnit](#)

UrlShortener · github.com/mdemrulkayes/UrlShortener

URL shortener on ASP.NET Core Minimal APIs with EF Core and PostgreSQL — short-code generation, redirects, and request validation; FluentValidation, OpenAPI + Scalar docs, and a MudBlazor UI calling the API via Refit. Tested with xUnit + Shouldly on the EF Core InMemory provider, shipped through a GitHub Actions CI pipeline.

[.NET 10](#) · [ASP.NET Core Minimal APIs](#) · [EF Core](#) · [PostgreSQL](#) · [Blazor Server](#) · [MudBlazor](#) · [Refit](#) · [FluentValidation](#) · [OpenAPI/Scalar](#) · [xUnit](#) · [GitHub Actions](#)

HotelManagementSystem · github.com/mdemrulkayes/HotelManagementSystem

Hotel management app over a shared Business/DataAccess/Common layering — an ASP.NET Core API with both Blazor Server and Blazor WASM clients, covering room creation, booking, and payment configuration; containerized via Docker Compose.

[.NET](#) · [ASP.NET Core API](#) · [Blazor Server](#) · [Blazor WASM](#) · [Layered Architecture](#) · [Docker Compose](#)

EDUCATION

M.Sc. in Computer Science — Jahangirnagar University

Dhaka, Bangladesh

B.Sc. in Computer Science & Engineering — Daffodil International University

Dhaka, Bangladesh